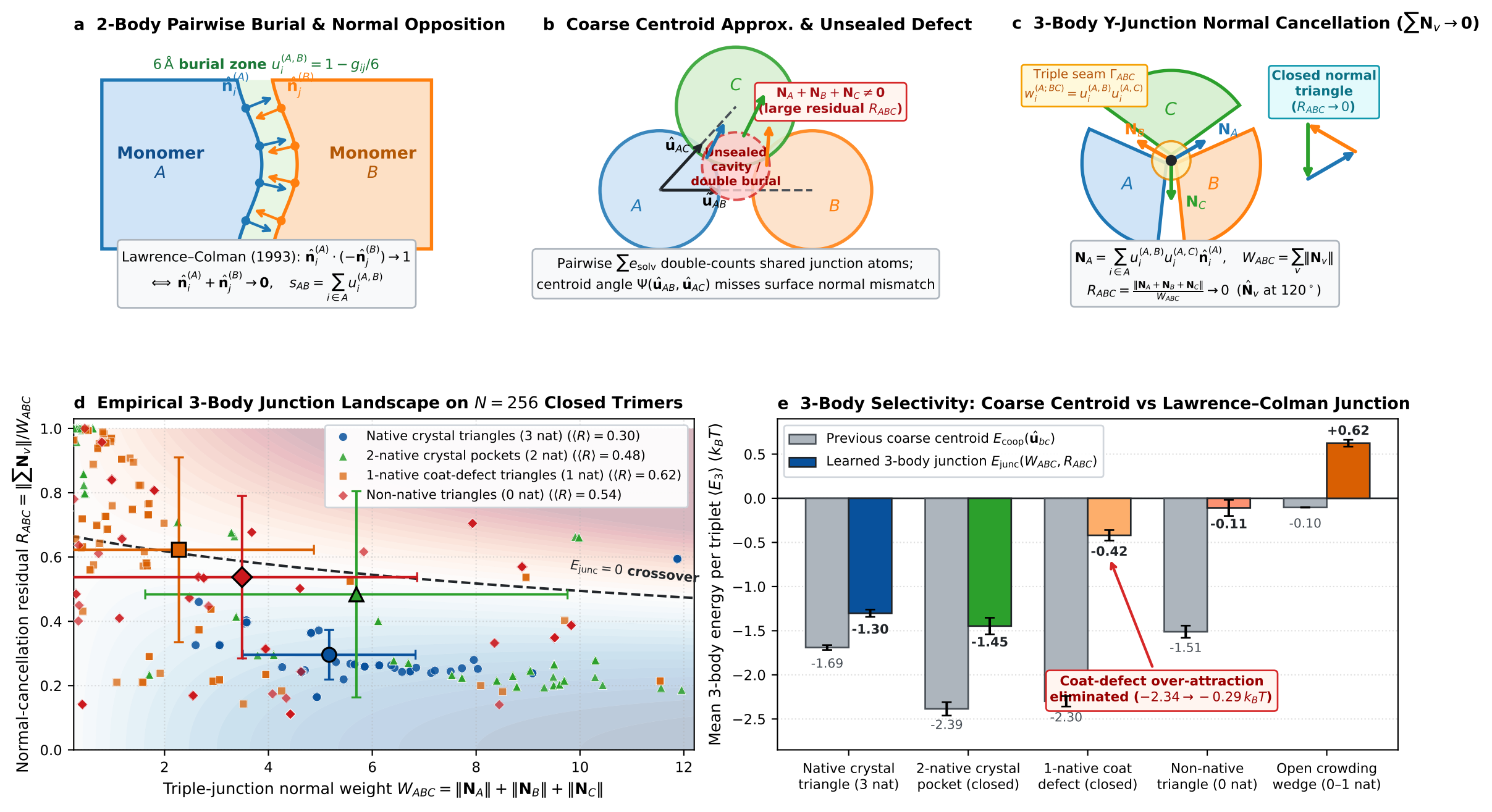

a 2-Body Pairwise Burial & Normal Opposition
6 Å burial zone $u_i^{(A,B)} = 1 - g_{ij}/6$
$\hat{\mathbf{n}}_i^{(A)}$
$\hat{\mathbf{n}}_j^{(B)}$
Monomer A
Monomer B
Lawrence–Colman (1993): $\hat{\mathbf{n}}_i^{(A)} \cdot (-\hat{\mathbf{n}}_j^{(B)}) \to 1$
$\iff \hat{\mathbf{n}}_i^{(A)} + \hat{\mathbf{n}}_j^{(B)} \to \mathbf{0}, \quad s_{AB} = \sum_{i \in A} u_i^{(A,B)}$
b Coarse Centroid Approx. & Unsealed Defect
C
$\mathbf{N}_A + \mathbf{N}_B + \mathbf{N}_C \neq \mathbf{0}$
(large residual R_{ABC})
$\hat{\mathbf{u}}_{AC}$
unsealed cavity / double burial
$\hat{\mathbf{u}}_{AB}$
A
B
Pairwise $\sum e_{solv}$ double-counts shared junction atoms;
centroid angle $\Psi(\hat{\mathbf{u}}_{AB}, \hat{\mathbf{u}}_{AC})$ misses surface normal mismatch
c 3-Body Y-Junction Normal Cancellation ($\sum \mathbf{N}_v \to 0$)
Triple seam Γ_{ABC}
$w_i^{(A;BC)} = u_i^{(A,B)} u_i^{(A,C)}$
Closed normal triangle
($R_{ABC} \to 0$)
C
$\mathbf{N}_B$
$\mathbf{N}_A$
A
B
$\mathbf{N}_C$
$\mathbf{N}_A = \sum_{i \in A} u_i^{(A,B)} u_i^{(A,C)} \hat{\mathbf{n}}_i^{(A)}, \quad W_{ABC} = \sum_v \|\mathbf{N}_v\|$
$R_{ABC} = \frac{\|\mathbf{N}_A + \mathbf{N}_B + \mathbf{N}_C\|}{W_{ABC}} \to 0$ ($\hat{\mathbf{N}}_v$ at 120°)
d Empirical 3-Body Junction Landscape on $N = 256$ Closed Trimers
Native crystal triangles (3 nat) ($\langle R \rangle = 0.30$)
2-native crystal pockets (2 nat) ($\langle R \rangle = 0.48$)
1-native coat-defect triangles (1 nat) ($\langle R \rangle = 0.62$)
Non-native triangles (0 nat) ($\langle R \rangle = 0.54$)
$E_{junc} = 0$ crossover
Normal-cancellation residual $R_{ABC} = \|\sum \mathbf{N}_v\| / W_{ABC}$
Triple-junction normal weight $W_{ABC} = \|\mathbf{N}_A\| + \|\mathbf{N}_B\| + \|\mathbf{N}_C\|$
e 3-Body Selectivity: Coarse Centroid vs Lawrence–Colman Junction
Previous coarse centroid $E_{coop}(\hat{\mathbf{u}}_{bc})$
Learned 3-body junction $E_{junc}(W_{ABC}, R_{ABC})$
Mean 3-body energy per triplet $\langle E_3 \rangle$ (k_BT)
-1.69
-1.30
-2.39
-1.45
-2.30
-0.42
-1.51
-0.11
-0.10
+0.62
Coat-defect over-attraction eliminated ($-2.34 \to -0.29\, k_BT$)
Native crystal triangle (3 nat)
2-native crystal pocket (closed)
1-native coat defect (closed)
Non-native triangle (0 nat)
Open crowding wedge (0–1 nat)